RAMCO INSTITUTE OF TECHNOLOGY
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Department of Computer Science and Engineering.
Academic Year: 2023 - 2024(Odd/~~Even~~ Semester)

Degree, Semester & Branch: B.E, V & CSE
Course Code & Title: CS3551& Distributed Computing
Name of the Faculty member (s): A. Sofia Thanga Mary, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit II/** Snapshot algorithms for FIFO channels
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 5
- **Activity Chosen:** Cross word puzzle
- **Justification:**
  To identify the understanding level of a student in Snapshot algorithm, a game based activity was chosen to do the revision of topic instead of carrying out regular quiz tests.

- **Time Allotted for the Activity: 10 Minutes**

- **Details of the Implementation:**
  - A crossword puzzle is a fun word game that encourages students to remember the idea and try to figure out the solution on their own. It encourages students' curiosity instead of giving them the typical MCQ exams.
  - The activity was conducted in a classroom for a duration of 10 minutes. The students were given 10 questions as a crossword puzzle in the topic of Snapshot algorithm. They have to think and write the answers in the given puzzle.
  - The activity was conducted during the last 20 minutes of the class after that the answers were corrected. Sample crossword puzzles solved by the students were shown in figure 1.
  - At the end of class, the correct answers to the questions were discussed to ensure that everyone understood the correct responses to all the questions.

- **Images / Screenshot of the practice:**

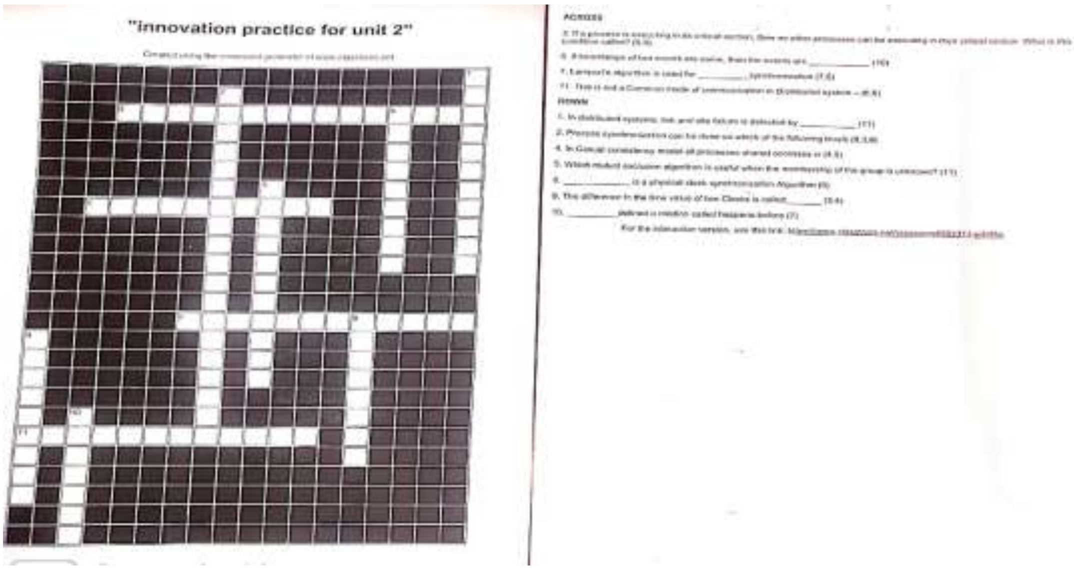

Fig 1: **crossword puzzle with question**

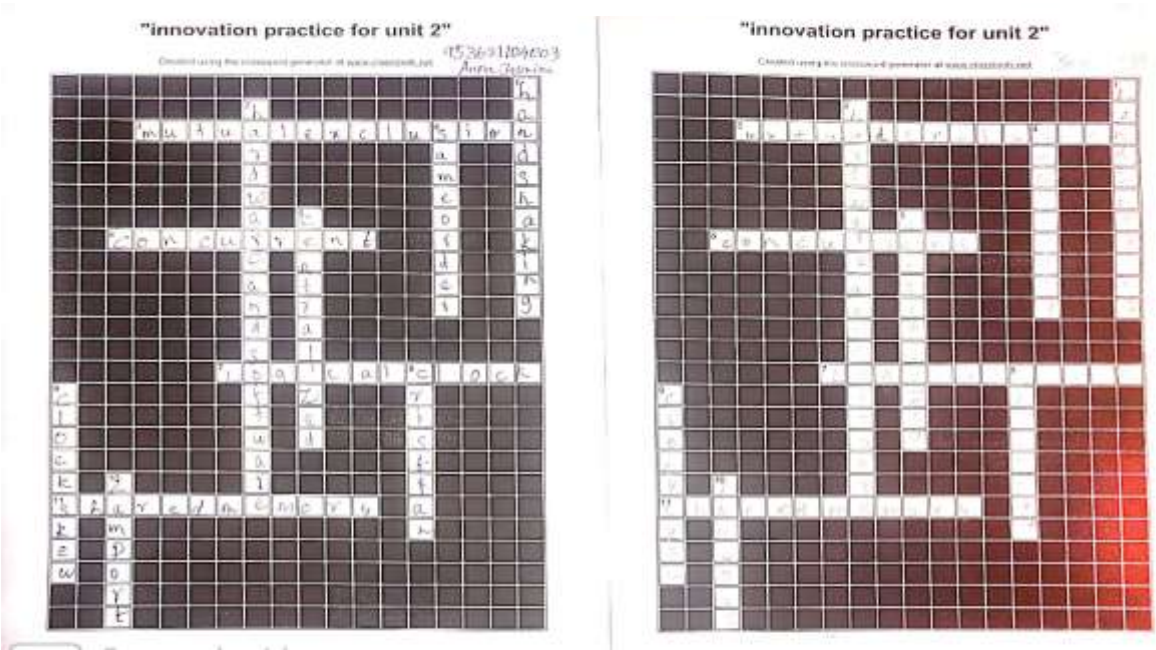

Fig 2: **sample answers**

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	1	2	2	2	3	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO4**	**PO8**
	3	**2**	**1**	**2**
Justification for correlation	Able to get the knowledge in group communication, global state and snapshot recording algorithm using Distributed system	Students will be able to identify, formulate and analyze the best solution with the basic knowledge of the global state and snapshot recording concepts.	Students will be able to use the knowledge of group communication, global state mechanism and Snapshot recording could be analyzed the problems in distributed computing.	The students could be able to do casual ordering and group communication by following the ethical principle or rules.

PSO1	**PSO2**
3	**3**
Students will be able to use the snapshot algorithms, global state and casual ordering mechanism to develop the software.	Students could be able to provide the distributed solution with the knowledge of message ordering and snapshot.

Reflective Critique:

- **Feedback of practice from students and other stakeholders:**
  - The crossword puzzle activity was very interesting and able to identify the appropriate name of the event handler.
  - It will be helpful while preparing for examination
- **Benefit of the practice:**
  - Student's understanding on event handler is improved.
  - Vocabulary of the terms related to event handling is improved
  - This activity helps to test the level of understanding of the students
- **Challenges faced in implementation:**
  - Few students found it difficult to complete the puzzle

- Students just might not have the necessary knowledge to complete crossword puzzles

❖ **References:**

- https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-GM.pdf
- http://www.classtools.net/crossword/
- https://www.activityvillage.co.uk/crosswords

Signature of Faculty Member **HOD**